relativity, and was impressed. That study led him to his 1908 discovery of the absolute nature of four-dimensional spacetime.

When Einstein learned of Minkowski's discovery, he was *not* impressed. Minkowski was merely rewriting the laws of special relativity in a new, more mathematical language; and, to Einstein, the mathematics obscured the physical ideas that underlie the laws. As Minkowski continued to extol the beauties of his spacetime viewpoint, Einstein began to make jokes about Göttingen mathematicians describing relativity in such complicated language that physicists wouldn't be able to understand it.

The joke, in fact, was on Einstein. Four years later, in 1912, he would realize that Minkowski's absolute spacetime is an essential foundation for incorporating gravity into special relativity. Sadly, Minkowski did not live to see this; he died of appendicitis in 1909, at age forty-five.

I shall return to Minkowski's absolute spacetime later in this chapter. First, however, I must develop another thread of my story: Newton's law of gravity and Einstein's first steps toward reconciling it with special relativity, steps he took before he began to appreciate Minkowski's breakthrough.

Newton's Gravitational Law, and Einstein's First Steps to Marry It to Relativity

Newton conceived of gravity as a force that acts between every pair of objects in the Universe, a force that pulls the objects toward each other. The larger the objects' masses and the closer they are together, the stronger the force. Stated more precisely, the force is proportional to the product of the objects' masses and inversely proportional to the square of the distance between them.

This gravitational law was an enormous intellectual triumph. When combined with Newton's laws of motion, it explained the orbits of the planets around the Sun, and the moons around the planets, the ebb and flow of ocean tides, and the fall of rocks; and it taught Newton and his seventeenth-century compatriots how to weigh the Sun and the Earth.[1]

During the two centuries that separated Newton and Einstein, astronomers' measurements of celestial orbits improved manyfold, putting